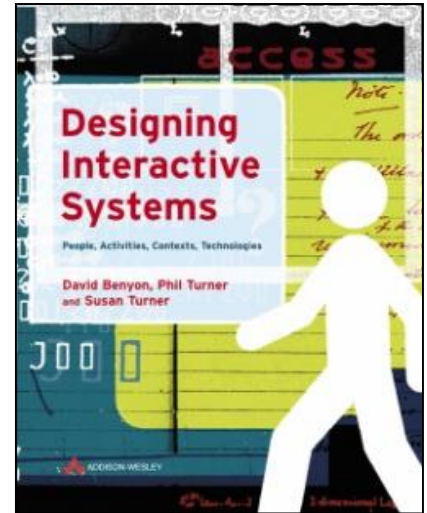


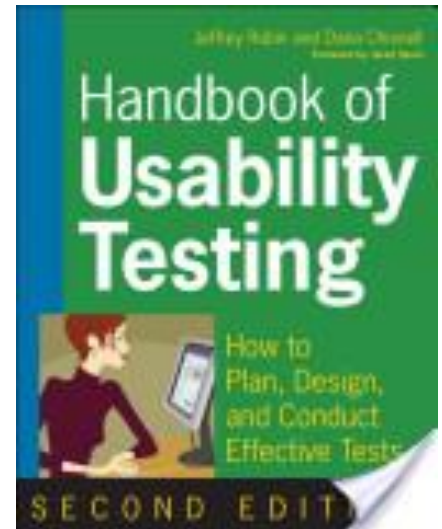
Usability Testing

Books

Designing Interactive Systems
Benyon, Turner and Turner



Handbook of Usability Testing
Rubin and Chisnell



Basics of the Methodology

- **A hypothesis must be formulated. A hypothesis states what you:** expect to occur when testing. For example, “Help as designed in format A will improve the speed and error rate of experienced users more than help as designed in format B.” It is essential that the hypothesis be as specific as possible.
- **Randomly chosen (using a very systematic method) participants must be assigned to experimental conditions. One needs to understand :** the characteristics of the target population, and from that larger population select a representative random sample. Random sampling is often difficult, especially when choosing from a population of existing customers.

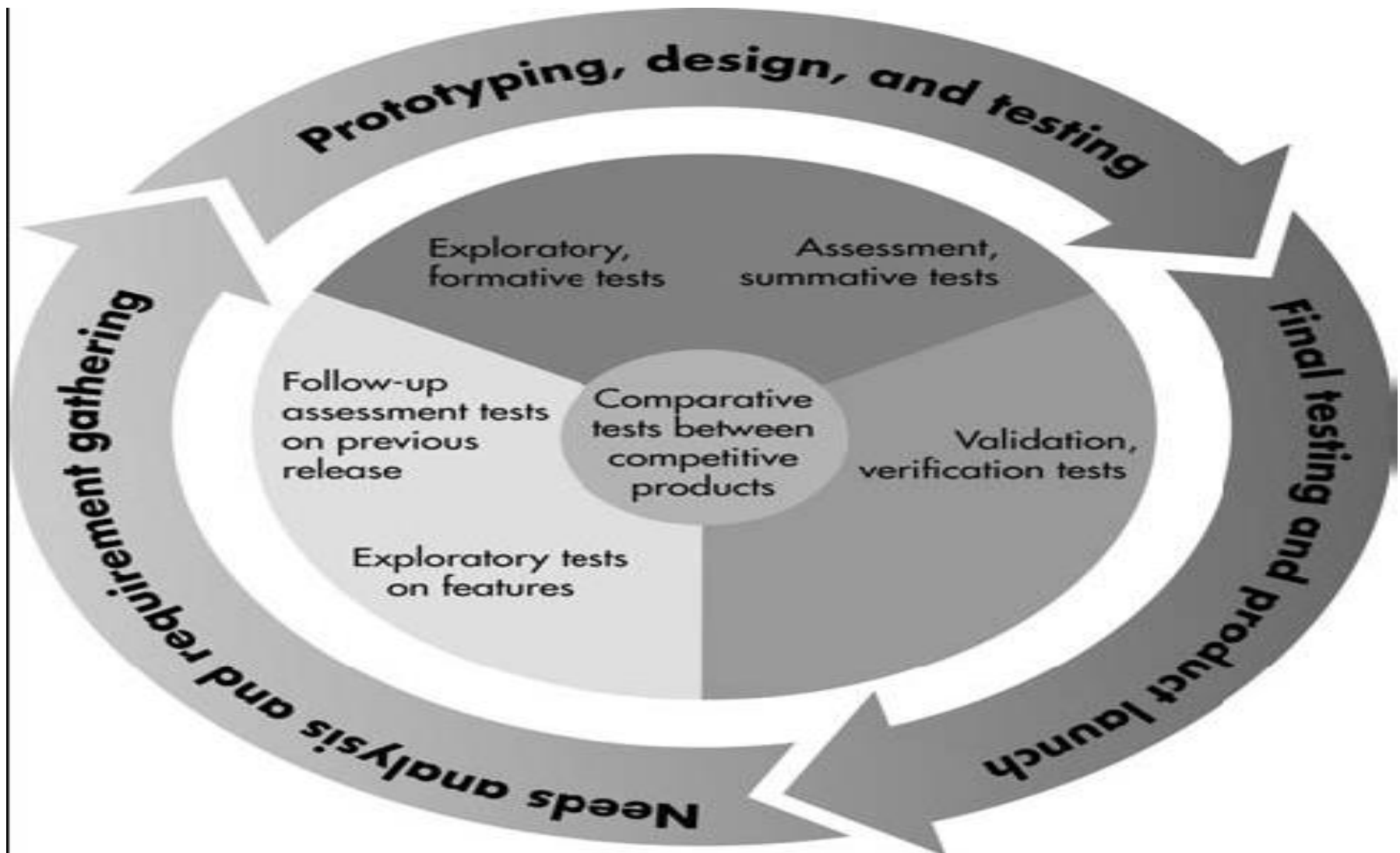
Basics of the Methodology

- **Tight controls must be employed. Experimental controls are crucial:** or else the validity of the results can be called into question, regardless of whether statistical significance is the goal. All participants should have nearly the identical experience as each other prior to and during the test. In addition, the amount of interaction with the test moderator must be controlled.
- **Control groups must be employed. In order to validate results, a control** group must be employed; its treatment should vary only on the single variable being tested.
- **The sample (of users) must be of sufficient size to measure statistically significant differences between groups. In order to measure differences** between groups statistically, a large enough sample size must be used. Too small a sample can lead to erroneous conclusions.

Basic Elements of Usability Testing

- ☐ Development of research questions or test objectives rather than hypotheses.
- ☐ Use of a representative sample of end users which may or may not be randomly chosen.
- ☐ Representation of the actual work environment.
- ☐ Observation of end users who either use or review a representation of the product.
- ☐ Controlled and sometimes extensive interviewing and probing of the participants by the test moderator.
- ☐ Collection of quantitative and qualitative performance and preference measures.
- ☐ Recommendation of improvements to the design of the product.

Usability testing throughout the product lifecycle



Exploratory or Formative Study

- **When**

The exploratory study is conducted quite early in the development cycle, when a product is still in the preliminary stages of being defined and designed (hence the reason it is sometimes called “formative”). By this point in the development cycle, the user profile and usage model (or task analysis) of the product will have (or should have) been defined. The project team is probably wrestling with the functional specification and early models of the product. Or perhaps the requirements and specifications phase is completed, and the design phase is just about to begin.

- **Objective**

The main objective of the exploratory study is to examine the effectiveness of preliminary design concepts. If one thinks of a user interface or a document as being divided into a high-level aspect and a more detailed aspect, the exploratory study is concerned with the former.

Exploratory or Formative Study

- The implications of these high-level issues go beyond the product, because you are also interested in verifying your assumptions about the *users*. *Understanding* one is necessary to define the other. Some typical user-oriented questions that an exploratory study would attempt to answer might include the following:
 - What do users conceive and think about using the product?
 - Does the product's basic functionality have value to the user?
 - How easily and successfully can users navigate?
 - How easily do users make inferences about how to use this user interface, based on their previous experience?
 - What type of prerequisite information does a person need to use the product?
 - Which functions of the product are “walk up and use” and which will probably require either help or written documentation?
 - How should the table of contents be organized to accommodate both novice and experienced users?

Exploratory or Formative Study

- Ask participants for their ideas about how to improve confusing areas.
- Unlike summative tests where there is more emphasis on measuring *how well the* user is able to perform by collecting quantitative data, here you strive to understand *why the user performs as he or she does by collecting qualitative* data. Regardless of whether you use a working prototype, static screens, early manuals, or whether the user performs tasks or simply “walks through” a product with the test moderator, the distinguishing feature of the exploratory test is its emphasis on discussion and examination of high-level concepts and thought processes, thus helping to form the final design.

Assessment or Summative Test

- **When**

The assessment test is probably the most typical type of usability test conducted. Of all the tests, it is probably the simplest and most straightforward for the novice usability professional to design and conduct. Assessment tests are conducted either early or midway into the product development cycle, usually after the fundamental or high-level design or organization of the product has been established.

- **Objective**

The purpose of the assessment test is to expand the findings of the exploratory test by evaluating the usability of lower-level operations and aspects of the product. If the intent of the exploratory test is to work on the skeleton of the product, the assessment test begins to work on the meat and the flesh. Assuming that the basic conceptual model of the product is sound, this test seeks to examine and evaluate how effectively the concept has been implemented. Rather than just exploring the intuitiveness of a product, you are interested in seeing how well a user can actually perform full-blown realistic tasks and in identifying specific usability deficiencies in the product.

Assessment or Summative Test

- It is often referred to as an information-gathering or evidence-gathering test, the methodology for an assessment test is a cross between the informal exploration of the exploratory test and the more tightly controlled measurement of the validation test. Unlike the exploratory test:
 - The user will always *perform tasks rather than simply walking through* and commenting upon screens, pages, and so on.
 - The test moderator will lessen his or her interaction with the participant because there is less emphasis on thought processes and more on actual behaviors.
 - Quantitative measures will be collected.

Validation or Verification Test

- **When**
- The validation test, also referred to as the verification test, is usually conducted late in the development cycle and, as the name suggests, is intended to measure usability of a product against established benchmarks or, in the case of a verification test, to confirm that problems discovered earlier have been remedied and that new ones have not been introduced. Unlike the first two tests, which take place in the middle of a very active and ongoing design cycle, the validation test typically takes place much closer to the release of the product.
- **Objective**
- The objective of the validation test is to evaluate how the product compares to some predetermined usability standard or benchmark, either a project-related performance standard, an internal company or historical standard, or even a competitor's standard of performance. The intent is to establish that the product meets such a standard prior to release, and if it does not, to establish the reason(s) why. The standards usually originate from the usability objectives developed early in the project. These in turn come from previous usability tests, marketing surveys, interviews with users, or simply educated guesses by the development team.

Validation or Verification Test

- The validation test is conducted in similar fashion to the assessment test with three major exceptions.
- Prior to the test, benchmarks or standards for the tasks of the test are either developed or identified. This can be specific error or time measures, or as simple as eliminating the problems identified in earlier exploratory tests.
- Participants are given tasks to perform with either very little or no interaction with a test moderator. (And they are probably not asked to “think aloud.”)
- The collection of quantitative data is the central focus, although reasons for substandard performance are identified.

Comparison Test

- **When**
- The comparison test is not associated with any specific point in the product development lifecycle. In the early stages, it can be used to compare several radically different interface styles via an exploratory test, to see which has the greatest potential with the proposed target population. Toward the middle of the lifecycle, a comparison test can be used to measure the effectiveness of a single element, such as whether pictorial buttons or textual buttons are preferred by users. Toward the end of the lifecycle, a comparison test can be used to see how the released product stacks up against a competitor's product.
- **Objective**
- The comparison test is the fourth type of test and can be used in conjunction with any of the other three tests. It is used to compare two or more designs, such as two different interface styles, or the current design of a manual with a proposed new design, or to compare your product with a competitor's. The comparison test is typically used to establish which design is easier to use or learn, or to better understand the advantages and disadvantages of different designs.

Comparison Test

- For exploratory comparison tests, experience has shown that the best results and the most creative solutions are obtained by including wildly differing alternatives, rather than very similar alternatives. This seems to work because:
- The design team is forced to stretch its conceptions of what will work rather than just continuing along in a predictable pattern. With the necessity for developing very different alternatives, the design team is forced to move away from predictable ways of thinking about the problem. Typically, this involves revisiting fundamental premises about an interface or documentation format that have been around for years. The result is often a design that redefines and improves the product in fundamental ways.
- During the test, the participant is forced to really consider and contemplate why one design is better and which aspects make it so. It is easier to compare alternatives that are very similar, but harder to compare very different ones. Why? Similar alternatives share the same framework and conceptual model, with only the lower-level operations working differently. Very different alternatives, however, are often based on different conceptual models of how each works and may challenge the user, especially one experienced with the product, to take stock of how the tasks are actually performed.

Example

- Read chapter 3 pages 32-44

test plan

- The test plan is the foundation for the entire test. It addresses the how, when, where, who, why, and what of your usability test.
- Under the sometimes unrelenting time pressure of project deadlines, there could be a tendency to forgo writing a detailed test plan. Perhaps, feeling that you have a good idea of what you would like to test in your head, you decide not to bother writing it down.

Test Plan

- Purpose, goals, and objectives of the test
- Research questions
- Participant characteristics
- Method (test design)
- Task list
- Test environment, equipment, and logistics
- Test moderator role
- Data to be collected and evaluation measures

Overall objectives for the study

We will gather baseline data about the overall effectiveness of H.com. The goals of this study are to:

- Assess the overall effectiveness of www.H.com for different types of users performing basic, common tasks.
- Identify obstacles to completing room reservations on the site.
- Create a repeatable usability study protocol.

Figure 5-1 Sample purpose and goals for a usability test

PRODUCT

Small
interfaces

RESEARCH QUESTIONS

How easily do users switch between modes on multi-purpose buttons?
How well do users understand the symbols and icons?
Which ones are problematic? Why?
How easily do users download updates and features?
How quickly can users perform common tasks?

PRODUCT

Web sites

RESEARCH QUESTIONS

How easily do users understand what is clickable?
How easily and successfully do users find the products or information they are looking for?
How easily and successfully do users register for the site?
Where in the site do users go to find Search? Why?
How easily can users return to the home page?

Characteristic	Desired number of participants
Participant type	
pilot	1
regular	12
backup	2
Total number of participants	14
Travel frequency	
<i>infrequently: 1–5 trips per year</i>	4
<i>moderately often: 6–12 trips per year</i>	4
<i>very often: 13 or more trips per year</i>	4
Types of travel	
mostly business	6
mostly leisure	6
Booking experience	
book their own trips and accommodations	all
book online most of the time	6
book on the phone or other method	6
Age	
21–30	2–3
31–40	4–5
41–50	4–5
51–60	2–3
Gender	
female	6
male	6

Figure 5-3 Sample participant characteristics and desired mix

Methodology

This usability study will be somewhat exploratory but will also gather assessment data about the effectiveness of www.H.com. Participants will fall into three groups by the starting point they use to perform the main task, which is to reserve a room. We will collect data about error and success rates as well as qualitative data about participants' experiences using the site.

We will use a between-subjects design

In this between-subjects study, each participant will work through one task path (in a within-subjects study, each participant would try all paths in counterbalanced order). I will conduct up to 30 individual 45-minute usability study sessions. Each participant will perform one of three major task "paths" using www.H.com. I'll use 15 minutes of each session to explain the session to the participant, review basic background information with the participant, and then conduct a post-test debriefing interview. During the middle 30 minutes of the session, participants will work to reserve a room at an H property in a major U.S. city.

Session outline and timing

The test sessions will be 45 minutes long. I will use 15 minutes of each session for pre-test introductions and post-test debriefing interviews. The sessions will take place at Shugoll Research in Bethesda.

Pre-test arrangements

Have the participant:

- Review and sign nondisclosures and recording permissions.
- Fill out a background questionnaire (with the same questions as the screener).

Introduction to the session (2 minutes)

Discuss:

- Participant's experience with usability studies and focus groups.
- Importance of their involvement in the study.
- Moderator's role.
- Room configuration, recording systems, observers, etc.
- The protocol for the rest of the session.
- Thinking aloud.

Background interview (3 minutes)

Discuss the participant's:

- Experiences booking their own travel.
- Reasons for booking their own travel.

Tasks (30 minutes)

Participants will start at one of three points to reserve a room at an H hotel in a major U.S. city where H has multiple properties.

Post-test debriefing (10 minutes)

- Ask broad questions to collect preference and other qualitative data.
- Follow up on any particular problems that came up for the participant.

Decide What Type of Information to Collect

Match the type of data you'll collect to a problem statement of the test plan. Figure 8-5 shows several matchups of problem statements with data collected.

Research Question	Data Collected
How easy is it to perform a specific task using a new hardware product?	Error rate among all participants. Or Number of steps required to perform the task.
Do novice participants access the site map?	Number of unsolicited accesses of the site map before completing the task. And Explanations of what participants were looking for in the navigation categories that they did not find.
How accessible is information in the Contact Us section of the web site?	Time expired between searching for a specific piece of information and locating it on the correct page.
How effective is a tutorial in teaching a specific task?	Time expired between accessing the tutorial and successful completion of the task. Or Comparison of error rates of participants using and not using the tutorial.

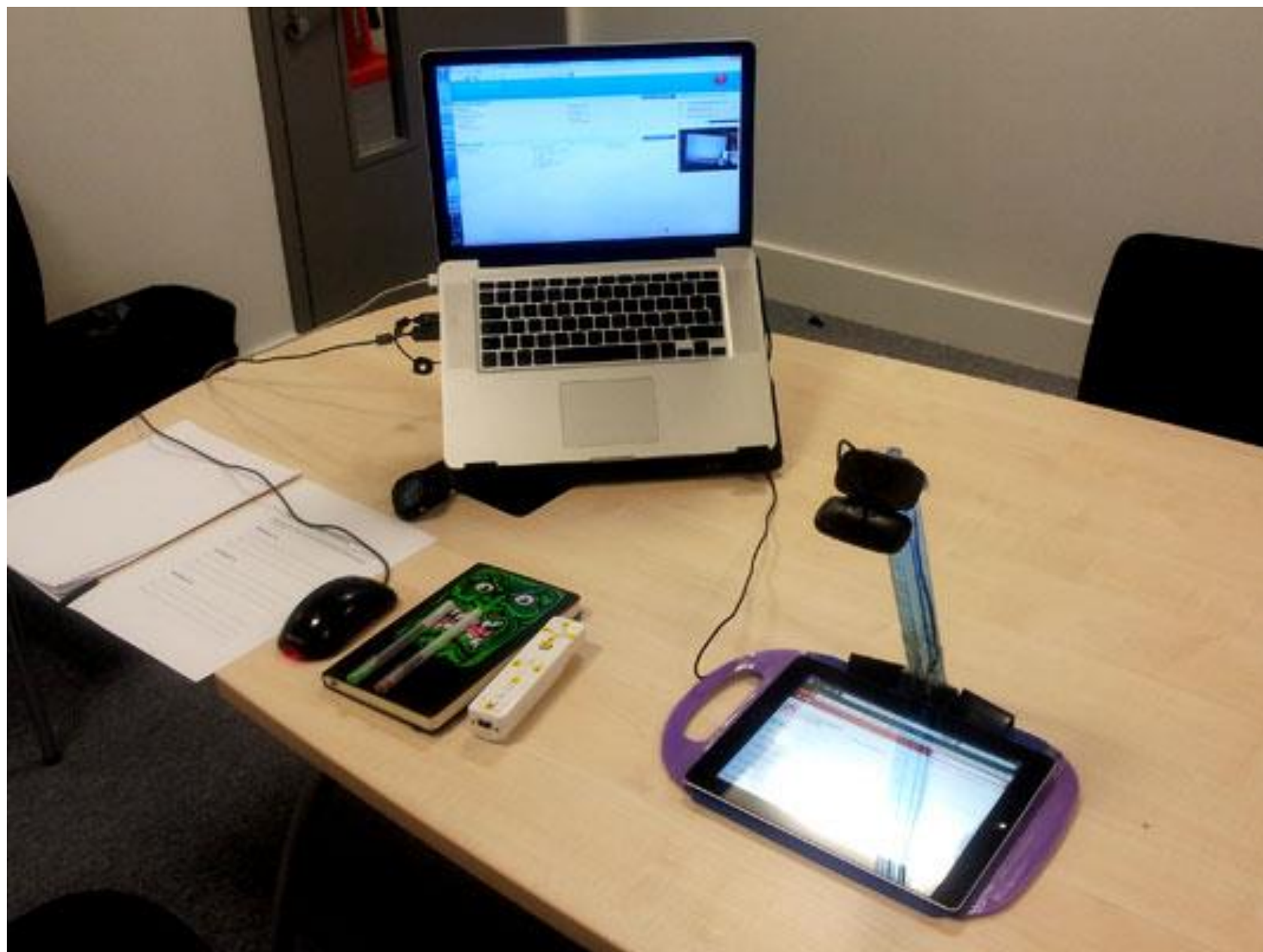
Figure 8-5 Research Question/Data Collection Table

- Chapters 8 and 9 have very useful details

Test Description

You need to explain properly

1. Goal of this observation study
2. Context and equipments
3. What he /she has to do
 - For example you need to ask him to
 1. call his daughter
 2. text his friend
 3. add a new friend





Data Collection and Analysis

- Measuring
 - Effective: completeness and accuracy
 - (number and type of Error)
 - Efficient: speed (
 - Time spent to in each interface and total time to complete each task

One Example

- Time spent to complete the tasks

Task	Elapsed Time	Notes
A	4:25	Clicked the wrong tab. Found his mistake. Clicked the wrong table again. Entered address incorrectly. Self-corrected after reading error message.
B	:30	Perfect. YES!
C	1:25	Very long hesitation. Pulls down the opportunities list and lean in close. Silence. Then selects and completes correctly. ??
D	2:30	Entered the wrong password. Very lost. "I hate it when that happens." Totally missed note after Step 8.

Figure 10-1 Data collection form showing probe location

Analysis and report results

- See chapter 11

Module	Tasks	Percentage of participants performing correctly (within benchmark)	Mean time (minutes)	Standard deviation (minutes)
A	Set temperature pressure	83	3.21	4.38
	Set gas flows	33	12.08	10.15 ¹
	Ignite the QRC	83	2.75	2.68
	Bake out column	100	0.46	0.17
	Set oven temperature program	66	6.54	2.56
	Run QRC checkout	100	0.83	0.34
	Program pressure and temperature	66	5.17	3.46
	Rerun checkout	100	0.29	0.09
B	Load the sample tray	100	0.88	0.28
	Create a subdirectory	33	8.00	3.92
	Create a sequence	66	9.42	7.70
	Start the sequence	66	1.00	0.92
	Stop the sequence	100	1.30	0.89
	Check the report	100 ²	1.38	0.58
	Reintegrate and print report	66	11.67	5.15
	Modify the method	100 ³	2.67	1.86
	Save the method	83	2.92	5.41
	Modify the sequence	66	3.46	3.96
	Restart the sequence	83	1.08	0.89

¹ Participants 4's timings are not included. Participant had great resistance to performing the task and had not attempted it on the job in more than 18 months. Participant was also given misinformation during the task.

² Participant 4 did not perform this task due to running the wrong method and having the test plot run.

³ Participant 1 did not perform this task to time constraints.

Figure 11-7 Non-criterion tasks shaded

Number of errors by version

	Version A	Version B	Version C
Novice users	6	6	15
Experienced users	9	20	12

Figure 11-5 Number of errors, compiled by version

Summary of performance and preference

Participant	Group	Version A # tasks correct	Version B # tasks correct	Liked best	Prefer to teach a novice	Version A Ease of use (1-5)	Version B Ease of use (1-5)
P1	N	*12/15	11/15	A	B	4	3
P2	N	12/15	*11/15	B	B	3	5
P3	N	*12/15	10/15	A	B	5	2
P4	N	10/15	*13/15	B	B	2	4
P5	E	*11/15	10/15	B	A	4	3
P6	E	11/15	*13/15	B	A	5	4
P7	E	*12/15	10/15	B	B	3	4
P8	E	13/15	*11/15	B	B	4	3
P9	N	*9/15	13/15	A	A	4	3
P10	N	12/15	*11/15	A	B	3	4
P11	E	*11/15	13/15	B	B	4	4
P12	E	14/15	*12/15	B	B	4	4
Avg.						3.8	3.6

Key:

N = Novice

E = Experienced

* = Participant saw this version first

Figure 11-6 Summary of performance and preference rankings

Severity ranking	Severity description	Severity definition
4	Unusable	The user either is not able to or will not want to use a particular part of the product because of the way that the product has been designed and implemented. Example: Product crashes unexpectedly whenever it is powered on at altitude.
3	Severe	The user will probably use or attempt to use the product, but will be severely limited in his or her ability to do so. The user will have great difficulty in working around the problem. Example: Synchronizing the device to another device can only happen when certain files are not in use. It isn't obvious when the files are in use.
2	Moderate	The user will be able to use the product in most cases, but will have to take some moderate effort in getting around the problem. Example: The user can make sure that all complementary applications are closed while syncing the two devices.
1	Imitant	The problem occurs only intermittently, can be circumvented easily, or is dependent on a standard that is outside the product's boundaries. Could also be a cosmetic problem. Example: The message area of the device's small screen is at the very top, dark blue, and often shaded by the frame of the screen.

Figure 11-9 Problem severity ranking

4 = Task failure—prevents this user going further

3 = Serious problem—may hinder this user

2 = Minor hindrance—possible issue, but probably will not hinder this user

1 = No problem—satisfies the benchmark

Figure 11-10 Problem severity ranking

Task	Source of error
Use points to reserve a hotel room	• Nothing in the web site said users must be logged in before starting the reservation process to be able to use points. • Participants expected to be able to sign in to see how many points they had after Step 3 (Check availability), but there was no way to log in there. • After the home page, the search widget that included an option to redeem points was in the lower-right corner of the page, and none of the participants noticed it. • Participants looked for a link to FAQs; they should have selected the Customer Service link, but participants inferred that the Customer Service link would lead to a phone number or online chat.

Figure 11-8 Excerpt from source of error analysis